

Technical Report: Provided-Text Conditioned Audio Delivery Ranking for EmpathyEval Track 1

WhereIsAI-Lingnan

Final Release, July 2026

Abstract

This report documents the final Phase 1 submission system, `provided_text_conditioned_audio_delivery_ranking`. The system uses a single `AudioDeliveryPairwiseMLP` trained on all official labeled training data. Candidate decisions use prosody, `Emotion2Vec`, `WavLM`, task, and text-length metadata only. ASR, teacher models, model fusion, fallback rules, speech-quality features, and test-time training are excluded. The frozen submission covers 530 questions and has been reproduced byte-for-byte from the released final checkpoint and retained final feature inputs.

1 Public Artifacts and Integrity

The public code repository is <https://github.com/Beichen-Hg/WhereIsAI-Lingnan>. The public Hugging Face model repository is <https://huggingface.co/BertramM/whereisai-lingnan-track1-audio>. The Hugging Face repository contains only the final model release files: `model.pt`, `config.yaml`, `feature_group_importance.json`, the model card, and a SHA-256 manifest. Competition data, cached feature files, pretrained-model copies, and per-question test predictions are not publicly released in those repositories.

Artifact	SHA-256
Final checkpoint <code>model.pt</code>	24303cb353b767c6f8c964876beec 85cbf868633aa88e3f291cf8004a5e23083
Checkpoint configuration <code>config.yaml</code>	d26c614e9eb27207bd6c9acdc00419e 64e998ec32899410e159564af1d9d5038
Frozen submission JSONL	f84f1a12568f2556db273fc3b70e7d87 b6319e83cf086c7772405e3abd185222

Table 1: Final release integrity identifiers.

2 Final Method

2.1 Model

The final learned model is `AudioDeliveryPairwiseMLP`. For each question, the scorer constructs candidate-level representations and converts them into pairwise features for ranking the available response audios. The checkpoint contains the model state dictionary, the 1,775 feature names, and the fitted feature normalization statistics needed by inference.

The exact feature groups are:

- **Prosody**: basic acoustic delivery statistics extracted by the `librosa_basic` backend.
- **Emotion2Vec**: audio emotion embeddings from `emotion2vec/emotion2vec_plus_large`.
- **WavLM**: speech embeddings from `microsoft/wavlm-base-plus`.
- **Task**: task-type indicators.
- **Text metadata**: lengths of official JSON-provided context, user, and response text.

Candidate semantic text is taken from the official JSON field `json_provided`. No ASR transcript is used to override provided text. FunASR is only an optional loading backend for the Emotion2Vec emotion model; it is not used as an automatic-speech-recognition decision module.

2.2 Excluded Components

The final inference path rejects or does not invoke ASR semantic replacement, teacher calls, test labels, fusion models, fallback models, speech-quality feature groups, or test-time optimization. Dependency and submission audits are run as part of final inference.

3 Training Protocol and Parameters

All official labeled training examples were used for the final refit. Phase 1 test examples were used only for blind manifest construction, feature extraction, inference, and submission auditing. No Phase 1 test labels are loaded by the final inference entry point.

Setting	Final value
Model	AudioDeliveryPairwiseMLP
Feature count	1,775
Hidden dimensions	[256, 128]
Dropout	0.15
Learning rate	0.0003
Weight decay	0.001
Batch size	128
Epochs	6
Random seed	2026
Training data	All official labeled training data

Table 2: Final refit configuration.

The all-labeled training metrics are retained as training diagnostics and are not claimed as held-out generalization results.

4 Runtime Environment

The final release was verified on Linux with Python 3.10.20, PyTorch 2.9.0+cu128, CUDA 12.8, NumPy 2.2.6, PyYAML 6.0.3, librosa 0.11.0, SoundFile 0.13.1, Transformers 5.8.1, Hugging Face Hub 1.14.0, FunASR 1.3.1, and ModelScope 1.36.3. GPU execution accelerates feature extraction;

the final scorer also supports `DEVICE=cpu`. A compatible PyTorch wheel must be selected for the target machine. Full audio-feature reconstruction may require system audio tooling such as `ffmpeg`, depending on the selected FunASR/torchaudio loading path.

5 Reproduction Procedure

The GitHub repository contains the final-only source code, scripts, tests, and runtime dependency specification. The Hugging Face checkpoint should be placed under the following project-relative directory:

```
artifacts/final_refit/audio_delivery_pairwise_all_labeled_clean_no_quality_v1/
```

With organizer-provided Phase 1 assets and the corresponding final feature files available under `artifacts/`, install dependencies and run:

```
python -m pip install -r requirements.txt
PYTHON_BIN=python DEVICE=cpu bash scripts/155_reproduce_final_submission.sh
```

The available scripts are:

- `140_build_phase1_full_provided_text_manifest.sh`: build the provided-text Phase 1 manifest.
- `141_build_phase1_full_provided_text_features.sh`: extract final prosody, Emotion2Vec, and WavLM features and build the final feature table.
- `142_infer_phase1_full_audio_delivery.sh`: execute final inference with the released checkpoint.
- `155_reproduce_final_submission.sh`: run inference and compare the generated JSONL byte-for-byte with the frozen candidate when that local audit file is available.

6 Validation and Submission Coverage

The final code tests passed 15/15. Final inference passed both the submission format audit and the dependency audit. The frozen submission contains 530 rows:

Source	Questions	Answer distribution
GigaSpeech	200	A=119, B=81
MELD	210	A=86, B=62, C=62
EmoV-DB	120	A=64, B=56
Total	530	A=269, B=199, C=62

Table 3: Frozen Phase 1 submission coverage.

7 Submission Checklist

- The Hugging Face model repository is public and ungated.
- The released checkpoint and configuration hashes are recorded above.
- The public GitHub repository contains code, runtime specifications, tests, and this report.
- The competition form should contain the public Hugging Face URL and the captain name required by the organizers.